

Observation

real LINEMOD RGB-D crop
+ 16 candidate grasps

Pose-then-plan

1. 6D pose estimator → pose (R, t)
2. → CAD mesh + simulator + physics prior

needs the object model and a physics simulator at decision time

Manipulation-sufficient perception

1. encoder $q_\theta(z \mid o)$ → code z (few bits)
2. → outcome head $g_\phi(z, a)$ for all candidates

trained on grasp outcomes; one payload scores every candidate

Outcome (simulated)



lifted dropped blocked

MuJoCo lift on the scanned mesh at the annotated pose; friction, mass and centre of mass drawn per scene, unseen